

FIFO

```
#include <stdio.h>
```

```
int main() {
```

```
    int pages[50], frames[10];
```

```
    int n, f, i, j;
```

```
    int pageFaults = 0;
```

```
    int next = 0;
```

```
    int found;
```

```
    printf("Enter number of pages: ");
```

```
    scanf("%d", &n);
```

```
    printf("Enter page reference string:\n");
```

```
    for(i = 0; i < n; i++) {
```

```
        scanf("%d", &pages[i]);
```

```
    }
```

```
    printf("Enter number of frames: ");
```

```
    scanf("%d", &f);
```

```
    for(i = 0; i < f; i++) {
```

```
        frames[i] = -1;
```

```
    }
```

```
    printf("\n--- FIFO Page Replacement ---\n");
```

```

for(i = 0; i < n; i++) {

    found = 0;

    for(j = 0; j < f; j++) {
        if(frames[j] == pages[i]) {
            found = 1;
            break;
        }
    }

    if(found == 0) {
        frames[next] = pages[i];
        next = (next + 1) % f;
        pageFaults++;
    }

    printf("Page %d -> [", pages[i]);

    for(j = 0; j < f; j++) {
        if(frames[j] != -1)
            printf("%d", frames[j]);
        else
            printf(" ");

        if(j != f - 1)
            printf(" ");
    }
}

```

```

        printf("\n");
    }

    printf("\nTotal Page Faults (FIFO): %d\n", pageFaults);

    return 0;
}

```

OUTPUT

```

Enter number of pages: 10
Enter page reference string:
1 2 0 4 3 2 1 1 3 4
Enter number of frames: 3

--- FIFO Page Replacement ---
Page 1 -> [1  ]
Page 2 -> [1 2  ]
Page 0 -> [1 2 0]
Page 4 -> [4 2 0]
Page 3 -> [4 3 0]
Page 2 -> [4 3 2]
Page 1 -> [1 3 2]
Page 1 -> [1 3 2]
Page 3 -> [1 3 2]
Page 4 -> [1 4 2]

Total Page Faults (FIFO): 8
PS C:\Users\osaun>

```

LRU

```
#include <stdio.h>
```

```

int main() {

    int pages[50], frames[10], time[10];

    int n, f, i, j;

```

```
int pageFaults = 0;

int counter = 0;

int found, pos, min;


printf("Enter number of pages: ");

scanf("%d", &n);


printf("Enter page reference string:\n");

for(i = 0; i < n; i++) {

    scanf("%d", &pages[i]);

}


printf("Enter number of frames: ");

scanf("%d", &f);


for(i = 0; i < f; i++) {

    frames[i] = -1;

    time[i] = 0;

}


printf("\n--- LRU Page Replacement ---\n");


for(i = 0; i < n; i++) {


    found = 0;


    for(j = 0; j < f; j++) {

        if(frames[j] == pages[i]) {
```

```
        found = 1;
        counter++;
        time[j] = counter;
        break;
    }
}
```

```
if(found == 0) {
```

```
    pos = -1;
```

```
    for(j = 0; j < f; j++) {
        if(frames[j] == -1) {
            pos = j;
            break;
        }
    }
}
```

```
if(pos != -1) {
    frames[pos] = pages[i];
    counter++;
    time[pos] = counter;
}
```

```
else {
```

```
    min = 0;
```

```
    for(j = 1; j < f; j++) {
        if(time[j] < time[min]) {
```

```

        min = j;
    }
}

frames[min] = pages[i];
counter++;
time[min] = counter;
}

pageFaults++;
}

printf("Page %d -> [", pages[i]);

for(j = 0; j < f; j++) {
    if(frames[j] != -1)
        printf("%d", frames[j]);
    else
        printf(" ");

    if(j != f - 1)
        printf(" ");
}

printf("]\n");
}

printf("\nTotal Page Faults (LRU): %d\n", pageFaults);

```

```
    return 0;
}
```

OUTPUT

```
Enter number of pages: 10
Enter page reference string:
1 2 3 4 0 0 2 3 2 1
Enter number of frames: 4

--- LRU Page Replacement ---
Page 1 -> [1      ]
Page 2 -> [1 2    ]
Page 3 -> [1 2 3   ]
Page 4 -> [1 2 3 4]
Page 0 -> [0 2 3 4]
Page 0 -> [0 2 3 4]
Page 2 -> [0 2 3 4]
Page 3 -> [0 2 3 4]
Page 2 -> [0 2 3 4]
Page 1 -> [0 2 3 1]

Total Page Faults (LRU): 6
PS C:\Users\osaun>
```

ORU

```
#include <stdio.h>
```

```
int main() {
    int pages[50], frames[10];
    int n, f, i, j, k;

    int pageFaults = 0;
    int found, pos, farthest, index;

    printf("Enter number of pages: ");
```

```
scanf("%d", &n);
```

```
printf("Enter page reference string:\n");
```

```
for(i = 0; i < n; i++) {
```

```
    scanf("%d", &pages[i]);
```

```
}
```

```
printf("Enter number of frames: ");
```

```
scanf("%d", &f);
```

```
for(i = 0; i < f; i++) {
```

```
    frames[i] = -1;
```

```
}
```

```
printf("\n--- Optimal Page Replacement ---\n");
```

```
for(i = 0; i < n; i++) {
```

```
    found = 0;
```

```
    for(j = 0; j < f; j++) {
```

```
        if(frames[j] == pages[i]) {
```

```
            found = 1;
```

```
            break;
```

```
        }
```

```
    }
```

```
    if(found == 0) {
```



```
pos = -1;
```

```
for(j = 0; j < f; j++) {  
    if(frames[j] == -1) {  
        pos = j;  
        break;  
    }  
}
```

```
if(pos != -1) {  
    frames[pos] = pages[i];  
}  
else {
```

```
    farthest = -1;  
    index = -1;
```

```
for(j = 0; j < f; j++) {
```

```
    int foundFuture = 0;
```

```
    for(k = i + 1; k < n; k++) {  
        if(frames[j] == pages[k]) {
```

```
            if(k > farthest) {  
                farthest = k;  
                index = j;
```

```
}
```

```
    foundFuture = 1;
```

```
    break;
```

```
}
```

```
}
```

```
if(foundFuture == 0) {
```

```
    index = j;
```

```
    break;
```

```
}
```

```
}
```

```
frames[index] = pages[i];
```

```
}
```

```
pageFaults++;
```

```
}
```

```
printf("Page %d -> [", pages[i]);
```

```
for(j = 0; j < f; j++) {
```

```
    if(frames[j] != -1)
```

```
        printf("%d", frames[j]);
```

```
    else
```

```
        printf(" ");
```

```

        if(j != f - 1)
            printf(" ");

    }

    printf("]\n");

}

printf("\nTotal Page Faults (Optimal): %d\n", pageFaults);

return 0;
}

```

OUTPUT

```

Enter number of pages: 10
Enter page reference string:
1 2 0 3 4 2 1 1 0 3
Enter number of frames: 3\

--- Optimal Page Replacement ---
Page 1 -> [1   ]
Page 2 -> [1 2  ]
Page 0 -> [1 2 0]
Page 3 -> [1 2 3]
Page 4 -> [1 2 4]
Page 2 -> [1 2 4]
Page 1 -> [1 2 4]
Page 1 -> [1 2 4]
Page 0 -> [0 2 4]
Page 3 -> [3 2 4]

Total Page Faults (Optimal): 7
PS C:\msys64\ucrt64\bin>

```